Measuring Logical Reasoning in Turkish Offline Language Models via Fallacy Detection

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***Abstract*—Logical fallacies are fundamental elements that cause the spread of disinformation and erroneous inferences within the digital information ecosystem. The ability of Large Language Models (LLMs) to detect these fallacies has become a critical benchmarking area for measuring the reasoning capacity of artificial intelligence. This study analyzes the performance of a local language model, `Ministral-3-14B-Reasoning`, in detecting and classifying Turkish fallacies using an approach that prioritizes data security. Within the scope of the study, an original Turkish dataset was developed consisting of more than 1000 examples, including five basic fallacy types (False Causality, Appeal to Authority, False Dilemma, Hasty Generalization, and Slippery Slope) and "contrast pairs" that are structurally similar to these fallacies but logically valid. Experimental results show that models exhibit limited reasoning in "Blind" scenarios where fallacy definitions are not provided (Accuracy: 0.611-0.640); however, when structured knowledge support is provided (Informed), success reaches 88.5% and the Cohen's Kappa value rises to a high level of 0.770. The findings demonstrate that local models have high potential in Turkish logical auditing processes and suggest that knowledge-augmented prompting can encourage System 2-like reasoning.**

***Index Terms*—Logical Fallacies, Large Language Models (LLMs), Local Language Models, Turkish Natural Language Processing (NLP), Reasoning Capacity, Knowledge-Augmented Prompting, Data Security.**

I. Introduction

The proliferation of digital communication environments has significantly increased the volume and variety of arguments individuals encounter; in parallel, the circulation of logical fallacies and deceptive reasoning has accelerated. Although logical fallacies appear superficially persuasive, they are arguments based on erroneous inferences that make it difficult to distinguish reliable information. Therefore, fallacy detection has become a fundamental research problem not only for critical thinking studies but also for automated text analysis systems. Specifically, the widespread adoption of Large Language Models (LLMs) has necessitated a systematic evaluation of not only their text generation capabilities but also their logical reasoning capacities.

Recent studies show that while LLMs excel at producing fluent and coherent text, they can yield inconsistent results in tasks requiring causal inference, generalization, and probabilistic reasoning [1]. The task of fallacy detection offers a particularly suitable testing ground for measuring the reasoning abilities of language models, as it requires multiple cognitive processes beyond superficial language patterns, such as causality, expertise evaluation, and statistical inference.

In the Turkish natural language processing literature, studies on logical fallacy detection are limited in number, and existing research is generally conducted on small-scale datasets [2]. This situation highlights the need for comprehensive datasets specifically designed for Turkish that represent different types of reasoning. Furthermore, in applications where data security is critical, evaluating the performance of local language models that do not depend on external API services has become an important requirement.

To address the dataset gap in Turkish literature and systematically evaluate the logical reasoning capacity of local language models, this study focuses on five fundamental fallacy types: *False Causality*, *Appeal to Authority*, *False Dilemma*, *Hasty Generalization*, and *Slippery Slope*. These fallacy types were selected to represent various reasoning processes, including causality, expertise evaluation, logical scope analysis, statistical generalization, and probabilistic inference. This allows for an analysis of not only the models' classification success but also their performance across different types of reasoning.

II. Related Work

The logical reasoning capabilities of artificial intelligence models constitute one of the most dynamic research areas in NLP literature in recent years. This section discusses current studies on fallacy detection, dataset creation methodologies, and the reasoning limitations of models.

A. Logical Reasoning and Limitations in LLMs

The reasoning processes of LLMs are often discussed in relation to Kahneman's dual-process theory of cognition [3]. According to this framework, System 1 refers to fast and intuitive judgments based on pattern recognition, whereas System 2 involves more deliberate and rule-oriented reasoning. Wang et al. [1] suggest that language models tend to operate in a System 1-like manner by default, while tasks such as fallacy detection require more structured analytical processing. Similarly, Payandeh et al. [4] demonstrate that models can be influenced by logically flawed arguments, highlighting limitations in robust reasoning.

Within this conceptual framework, the two evaluation settings used in this study can be interpreted as different levels of reasoning support. The "Blind" condition reflects model behavior without explicit guidance, while the "Informed" condition introduces structured fallacy definitions that encourage more deliberate analysis. This distinction provides a useful lens for interpreting performance differences between the two evaluation modes.

B. Fallacy Detection Datasets and Methodologies

One of the most comprehensive datasets in the literature, COCOLOFA [5], was created using LLM-supported crowdsourcing and measures the success of models in detecting fallacies in news comments. Li et al. [6] proposed the LFUD dataset, which aims for models not just to recognize fallacies but to understand "why" they are fallacies (Reason from Fallacy). Teo et al. [7] presented a comparative performance of different LLM families (LLaMA, Qwen, Gemma) in fallacy detection using the RAFDA framework.

C. Use of LLMs in Academic Writing and HCI Research

Research on incorporating LLMs into critical thinking processes warns that these systems should not be viewed as a "panacea." Lim and Perrault [8] state that the use of LLMs in HCI research requires rigorous evaluation. In this context, the study conducted by Yilmaz et al. established a critical reference point for model selection by examining the pedagogical and logical consistency of models' responses to absurd or misleading questions.

D. The Landscape in Turkish Literature

Specifically for Turkish, Uyar [2] analyzed the success of the GPT-3.5 model in detecting informal fallacies and found that the model performed poorly on certain fallacy types. Our study aims to advance research in this field by using a more up-to-date and local model (Ministral-3-14B-Reasoning) and by providing a large-scale dataset containing "fallacy-valid argument" contrast pairs, which are currently lacking in the literature.

III. Methodology

This study is designed to measure the ability of a local LLM to detect and classify Turkish logical fallacies. The methodology consists of three main stages: dataset creation, model selection, and experimental procedure.

A. Dataset Creation (TR-Fallacy Dataset)

The dataset, which is one of the primary contributions of this study, was developed using a hybrid approach. Considering the effectiveness of LLM-supported crowdsourcing methods in the literature [5], the following steps were followed:

1) **Human-Sourced Data:** A total of 150 original examples were produced by three experts in the field, covering five basic fallacy types (False Causality, Appeal to Authority, False Dilemma, Hasty Generalization, and Slippery Slope) along with arguments that are structurally similar to these fallacies but logically valid.
2) **Synthetic Data Generation:** Over 1000 synthetic examples were generated using various popular language models (Gemini, ChatGPT, Claude, Copilot, Qwen, DeepSeek). Half of these examples contain fallacy categories, while the other half consist of valid arguments referred to as "contrastive pairs."
3) **Categorization:** The dataset was labeled to be suitable for both binary (valid/fallacy) and multi-class (fallacy type) classification.

B. Model Selection: Ministral-3-14B-Reasoning

In selecting the model, findings from the study by Yilmaz et al. [9], which examines the pedagogical and logical responses of models to absurd questions, were taken as a reference. The `Ministral-3-14B-Reasoning` model was preferred based on criteria such as local (offline) usability, speed, data security, and reasoning success in the Turkish language. This model was specifically chosen due to its high scores in complex reasoning processes (Chain-of-Thought) and speed.

C. Experimental Procedure and Evaluation Methodology

The performance of the models was tested in two different stages:

1) Blind Evaluation (Zero-shot Detection): In this stage, texts were presented directly to the model without providing any fallacy definitions or examples. The model was asked to perform the following two tasks:

- **Binary Classification:** Determining whether the text is a fallacy or a valid argument.
- **Multi-class Identification:** Classifying which fallacy type or valid argument category the text is closer to.

2) Prompted Evaluation: In the second stage, summary definitions of the five selected fallacy types were added as a system prompt. This stage was designed to measure how effectively the model can integrate its internal knowledge with the provided definitions.

D. Transparency and Reproducibility

In accordance with the principle of scientific transparency, all prompts used in dataset production, raw human-generated data, and the Python codes used in the evaluation phase have been made available to researchers via an open-source GitHub repository[1]. This structure ensures the reproducibility of the study for similar languages and different model families.

IV. Results and Analysis

In this section, the performance of the `Ministral-3-14B-Reasoning` model in Turkish fallacy detection and classification tasks is analyzed across the axes of data source (Human/AI) and evaluation method (Blind/Informed).

A. Binary Classification Performance (Fallacy vs. Valid)

Binary classification results indicate that the model's capacity to distinguish logical validity is directly dependent on the prompting strategy (Table I).

The findings reveal that the model is significantly more successful in the "Informed" scenario (where fallacy definitions are provided). Specifically, the increase in accuracy from 0.611 to 0.885 in AI-generated data shows that the model gains high sensitivity in recognizing synthetic logical errors when structural definitions are provided. The increase in Cohen's Kappa values ($0.280 \rightarrow 0.656$) proves that the success achieved in the "Informed" stage is not coincidental

[1]Materials available at: github.com/kahramankostas/TRfallacy/

and that the model's reasoning process has stabilized. The performance differences between Blind and Informed conditions can be visually observed in Fig. 1, where the reduction of misclassifications is evident.

B. Multi-class Identification and Fallacy Type Analysis

When the model's ability to distinguish fallacy types is examined, significant performance differences were observed in certain categories:

- **Human Data:** The accuracy rate, which was 0.580 in the blind evaluation, rose to 0.700 with the inclusion of definitions.
- **AI Data:** In multi-class classification, the accuracy rate increased from 0.781 (Blind) to 0.830 (Informed).

This situation parallels the "System 1 vs. System 2" distinction proposed by Wang et al. [1]. While the model focuses on superficial similarities without definitions (System 1), it can perform a more in-depth logical analysis when definitions are provided (System 2). The improvements in fallacy-type discrimination are illustrated in Fig. 2, which shows reduced inter-class confusion under the Informed condition.

C. Discussion

The results of the study demonstrate that the local model, Ministral-3-14B-Reasoning, can exhibit highly competitive performance in complex reasoning tasks such as fallacy detection. However, the low performance in the "Blind" scenario confirms that models' internalized knowledge of fallacies is still limited or that they require structured guidance to trigger this knowledge.

The particular difficulty in detecting human-generated fallacies may stem from semantic ambiguity in natural language and cultural context. This finding supports the results reported by Uyar [2] on GPT-3.5, indicating that models struggle with informal fallacies. On the other hand, the high success rate in AI-generated data (0.885 Accuracy) suggests that synthetic data may carry more distinct structural traces and that models are optimized to recognize these traces.

An interesting anomaly appears in the Human-Informed condition, where accuracy increases substantially while macro precision and F1 scores decrease. This effect can be explained by the model's increased tendency to classify arguments as fallacies when definitions are provided.

Inspection of the confusion matrix suggests a rise in false positive classifications, indicating that structured definitions improve fallacy sensitivity but reduce discrimination between fallacious and valid arguments.

This phenomenon may indicate that knowledge-augmented prompting encourages cautious reasoning behavior, where the model prefers to detect potential fallacies rather than risk missing them. Similar behavior has been observed in risk-sensitive classification tasks and suggests that increased reasoning guidance does not necessarily improve all evaluation metrics simultaneously.

Consequently, the use of "knowledge-augmented" approaches in fallacy detection systems is a critical requirement for local models to operate securely and accurately.

D. Limitations

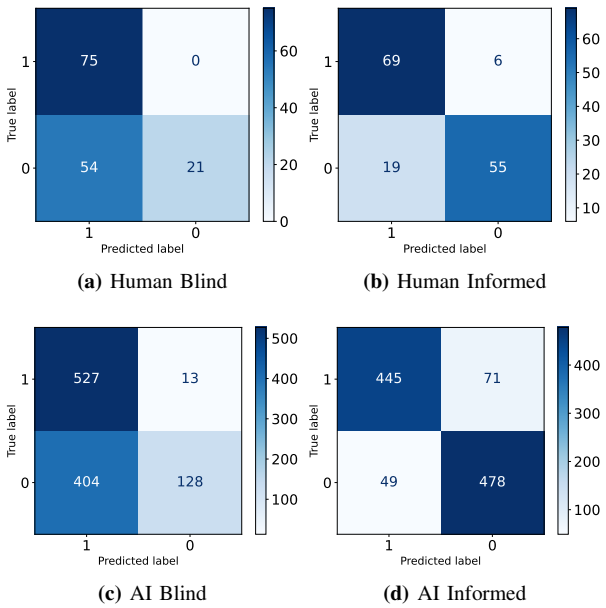
A key limitation of this study concerns the potential confounding effect between fallacy detection and stylistic pattern recognition. Since a substantial portion of the dataset was generated using LLMS, the evaluated model may have partially learned to distinguish AI-generated writing patterns rather than purely identifying logical fallacies.

This possibility is supported by the substantially higher performance observed on AI-generated examples compared to human-written arguments. While synthetic data enables large-scale dataset construction, it may introduce stylistic regularities that are easier for models to recognize than genuine logical structures.

Another limitation is the restricted scope of fallacy types included in the dataset. The analysis focuses on five commonly observed fallacies selected to represent different reasoning patterns. More complex fallacies, particularly context-dependent or dialogue-based errors, remain outside the scope of the present study.

Future work should therefore incorporate larger proportions of naturally occurring arguments and cross-domain evaluations to ensure that fallacy detection performance reflects reasoning ability rather than stylistic pattern matching. The relatively lower performance observed on human-written arguments highlights the challenges posed by natural language ambiguity and cultural context.

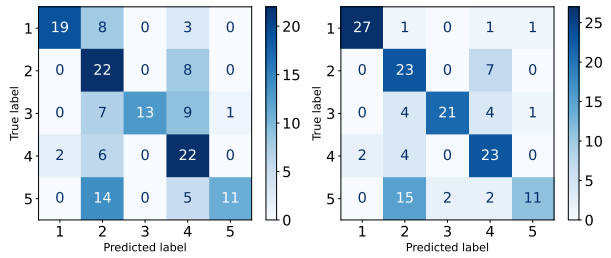
Finally, the experiments were conducted using a single local language model. Although the selected model provides a useful case study for offline reasoning systems, comparative evaluations across different model architectures and parameter scales would provide a more comprehensive understanding of fallacy detection capabilities.



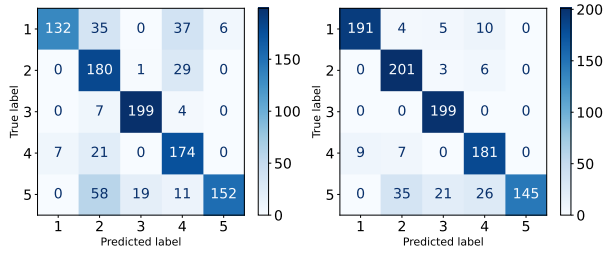
(a) Human Blind **(b)** Human Informed

(c) AI Blind **(d)** AI Informed

Fig. 1: Comparison of confusion matrices for binary classification (Fallacy vs. Valid). The matrices illustrate the performance shift between "Blind" and "Informed" evaluation modes across both human-sourced and AI-generated datasets, highlighting the model's improved sensitivity in the prompted scenario. 1: Fallacy, 2: Valid



(a) Human Blind **(b)** Human Informed



(c) AI Blind **(d)** AI Informed

Fig. 2: Confusion matrices for multi-class identification of specific fallacy types. The results demonstrate the model's ability to categorize different reasoning errors and reveal the reduction in inter-class misclassifications when structured definitions are integrated into the system prompt. 1: Appeal to Authority, 2: False Cause, 3: False Dilemma, 4: Hasty Generalization, 5: Slippery Slope.

TABLE I: Results of the detection of human-generated and AI-generated sentence classifications by the offline language model

		HUMAN		AI	
		Blind	Informed	Blind	Informed
Binary	Accuracy	0.640	0.827	0.611	0.885
	Precision (macro)	0.791	0.562	0.737	0.886
	Recall (macro)	0.640	0.551	0.608	0.885
	F1 (macro)	0.586	0.552	0.548	0.885
	Cohen Kappa	0.280	0.656	0.218	0.770
	MCC	0.403	0.666	0.320	0.770
Multi Metrics	Accuracy	0.580	0.700	0.781	0.879
	Precision (macro)	0.735	0.634	0.820	0.891
	Recall (macro)	0.580	0.583	0.786	0.885
	F1 (macro)	0.590	0.584	0.783	0.877
	Cohen Kappa	0.475	0.626	0.727	0.849
	MCC	0.499	0.638	0.736	0.854

V. Conclusion and Future Work

In this study, the Turkish logical fallacy detection and classification capabilities of a local LLM, `Ministral-3-14B-Reasoning`, were analyzed comprehensively. The hybrid dataset containing over 1000 examples (human and AI-generated) created within the scope of this work provides one of the largest open-source resources in this field to the Turkish natural language processing literature.

The findings prove that local models can be utilized for critical logical analysis tasks while preserving data security and privacy. However, it was observed that model performance is directly related to structured knowledge support (informed prompting). Specifically, the attainment of 88.5% binary classification accuracy in the "Informed" scenario and the rise of the Cohen's Kappa value from 0.218 to 0.770 indicate that models can transition from "System 1" intuitiveness to "System 2" depth when logical rules are provided.

Another significant outcome of the study is that models are more successful at detecting AI-generated fallacies compared to human-sourced examples. This situation reveals that semantic ambiguity and cultural nuances in human language remain a threshold that local models have yet to fully overcome.

A. Future Work

The following directions are suggested for future research:

- **Hybrid Reasoning Architectures:** Increasing the precision of fallacy detection by integrating local models with symbolic logic solvers (neuro-symbolic).
- **Expanded Fallacy Taxonomy:** Incorporating more complex fallacy types requiring dialogue context, such as "Red Herring" or "Straw Man," into the dataset.
- **Model Size and Quantization Analysis:** Comparing the effects of different parameter sizes and quantization techniques on the reasoning capacity of local models.

In conclusion, this study has laid the groundwork for a secure, adaptive, and high-performance logical auditing system for Turkish and validated the potential of local models in critical thinking processes.

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